

Melody-to-Piano: Transformer-Based Full Piano Accompaniment Generation from Simple Melody

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Abstract

This project studies melody-conditioned symbolic music generation in the setting of automatic piano accompaniment prediction. Given a monophonic melody sequence, the objective is to generate a structured polyphonic piano accompaniment containing harmonic compatibility, rhythmic support, and left- and right-hand register organization. The task is formulated as a conditional sequence-to-sequence learning problem over event-based MIDI tokens.

During the milestone stage, we implemented a complete experimental pipeline including symbolic preprocessing of the POP909 dataset, segmentation into fixed-length phrases, a deterministic rule-based accompaniment baseline, Transformer encoder-decoder models, and objective symbolic evaluation metrics. Preliminary experiments show that explicit chord conditioning improves note-level F1 from 0.0676 to 0.0712, onset-level F1 from 0.1618 to 0.1740, chord-tone ratio from 0.4772 to 0.5168, and register balance error from 0.1333 to 0.1201. These results suggest that harmonic context improves accompaniment prediction quality, while also showing that accurate polyphonic realization remains difficult under the current modeling setup. Remaining work will focus on decoding improvements, hyperparameter tuning, and deeper qualitative analysis.

1 Problem Statement and Motivation

Automatic accompaniment generation from melody is a structured symbolic prediction problem. Unlike unconditional music generation, accompaniment prediction must remain compatible with an input sequence while producing musically plausible output. In the piano setting, this requires simultaneous modeling of harmonic support, temporal alignment, chord density, and register allocation across left- and right-hand ranges.

This project studies melody-to-piano generation as a conditional sequence modeling task. A melody token sequence serves as the source signal and a polyphonic accompaniment sequence serves as the target output. The overall objective is to determine whether attention-based sequence models can learn accompaniment structure directly from symbolic datasets and whether explicit harmonic conditioning improves generation quality.

This problem is interesting for two reasons. First, it provides a controlled setting for studying structured generation with weak alignment between input and output. The melody constrains harmonic and rhythmic possibilities, but the accompaniment remains multimodal: multiple valid accompaniments may exist for the same melody. Second, melody-conditioned accompaniment is practically useful in composition tools, symbolic arrangement systems, and educational software where lead melodies are more easily available than full piano realizations.

The central research question is therefore: can a Transformer-based sequence-to-sequence model generate harmonically coherent and rhythmically structured piano accompaniment from melody input, and does explicit chord conditioning improve performance?

2 Related Work

Symbolic music generation has traditionally been studied using recurrent neural networks and autoregressive sequence models. Early work on neural music modeling used RNNs and LSTMs to represent temporal dependencies in pitch and rhythm sequences. These models demonstrated the feasibility of neural symbolic generation, but they were limited in representing long-range structure and hierarchical repetition patterns (Eck and Schmidhuber, 2002; Hadjeres and Pachet, 2017).

Transformer-based architectures later became

dominant because self-attention provides direct access to long contextual windows. Music Transformer introduced relative positional attention and showed that long symbolic music sequences can be modeled more effectively when the architecture explicitly captures relative timing relationships (Huang, 2019). This is particularly relevant for accompaniment generation, where harmonic context may span several measures.

Another relevant line of work studies multi-track symbolic generation and structured conditioning. MuseGAN explored generation of multiple coordinated symbolic tracks and highlighted the difficulty of preserving inter-track coherence (Dong, 2018). Although our task is simpler than full multi-track generation, accompaniment modeling still requires the system to learn polyphonic texture conditioned on an external melodic signal.

The POP909 dataset provides aligned melody, accompaniment, beat, and chord annotations specifically designed for symbolic arrangement research (Wang, 2020). This makes it particularly suitable for accompaniment learning because it exposes both melodic structure and explicit harmonic labels. In contrast to unconditional generation and continuation tasks, the present project focuses on melody-conditioned accompaniment prediction and uses chord conditioning as an explicit structural prior.

3 Approach

3.1 Task Formulation

Let $M = (m_1, \dots, m_T)$ denote the melody token sequence and $P = (p_1, \dots, p_S)$ denote the accompaniment token sequence. The task is modeled as learning a conditional distribution

$$P_\theta(P \mid M),$$

where P_θ is implemented as a Transformer encoder-decoder model trained with token-level cross-entropy loss under teacher forcing.

3.2 Symbolic Representation

MIDI tracks are converted into event-based symbolic tokens. The representation includes note on-set, note offset, temporal shift, and velocity events. A phrase-level segmentation step is used to divide long songs into fixed-length examples of 4–8 bars. This reduces sequence length variability and makes batch training more stable.

Only two tracks are used for the main task formulation:

- **MELODY**: used as the input sequence
- **PIANO**: used as the accompaniment target

The **BRIDGE** track is excluded from the current formulation because it acts as an auxiliary melodic layer rather than the main accompaniment texture. Including it would make the target less well-defined and introduce an additional melodic signal into what is intended to be a melody-to-accompaniment mapping.

3.3 Model Variants

Three systems are studied.

Rule-Based Baseline. A deterministic accompaniment generator uses chord labels to produce simple accompaniment patterns. The purpose of this baseline is not to achieve high realism, but to provide a reproducible symbolic reference that uses explicit harmonic information without neural learning.

Melody-Conditioned Transformer (Mel-Transformer). This system uses a Transformer encoder-decoder in which the encoder receives melody tokens only. The decoder predicts accompaniment tokens autoregressively.

Chord-Conditioned Transformer (Chord-Transformer). This variant augments the encoder input with chord tokens. Chord information is appended to the melody representation to provide explicit harmonic context. The comparison between the two neural models isolates the effect of harmonic conditioning.

3.4 Implementation Scope

The main project-specific implementation work includes:

- MIDI inspection and track extraction for POP909
- symbolic tokenization and segmentation pipeline
- chord parsing and conditioning input construction
- rule-based accompaniment baseline
- evaluation code for symbolic metrics

Property	Melody	Piano
Avg. notes	~320	~980
Texture	mono	poly
Pitch span	narrow	wide
Role	input	target

Table 1: Observed symbolic characteristics after pre-processing.

- training and decoding scripts for melody-conditioned and chord-conditioned models

Standard PyTorch Transformer modules are used for the neural architecture itself, while task-specific preprocessing and evaluation logic were implemented specifically for this project.

4 Experimental Setup

4.1 Dataset Analysis

Experiments use the POP909 dataset, which contains 909 pop songs with aligned symbolic annotations. After symbolic conversion, the songs are split into training (80%), validation (10%), and test (10%) partitions.

The dataset exhibits a clear difference between melody and accompaniment tracks. Melody lines are sparse, largely monophonic, and concentrated in a narrower pitch band. In contrast, piano accompaniment tracks contain substantially denser note activity and span a wider pitch range, reflecting the presence of bass notes, inner voices, and chord tones.

These observations are important because they show that accompaniment prediction is not a trivial copying problem. The model must learn to generate a denser symbolic structure than the input melody while remaining harmonically compatible with it.

4.2 Training Setup

Both neural models are trained using the Adam optimizer with early stopping based on validation loss. The melody-conditioned model reached its best validation loss of 1.0854, while the chord-conditioned model improved further to 1.0770. Training was stable in both cases, and the chord-conditioned model converged slightly later than the melody-only model.

The current experiments use greedy decoding during evaluation. This choice keeps the decoding procedure simple and reproducible, but it likely contributes to sparse outputs and lower recall,

since locally optimal token decisions can prevent better long-range accompaniment structure from emerging.

4.3 Evaluation Metrics

Prediction quality is measured using symbolic event-level metrics.

Note Precision and Recall. Let N_{pred} denote the set of predicted notes and N_{ref} denote the set of reference notes. Precision and recall are defined as

$$P = \frac{|N_{\text{pred}} \cap N_{\text{ref}}|}{|N_{\text{pred}}|}, \quad R = \frac{|N_{\text{pred}} \cap N_{\text{ref}}|}{|N_{\text{ref}}|}.$$

The note-level F1 score is

$$F_1 = \frac{2PR}{P + R}.$$

Onset Precision, Recall, and F1. The same formulation is applied to onset events only. These metrics ignore note duration and therefore focus more directly on rhythmic placement.

Chord-Tone Ratio. Harmonic consistency is measured using chord-tone ratio. Let G denote the set of generated notes and let $C(t_g)$ denote the set of pitch classes belonging to the chord active at the onset time of note g . Then

$$CTR = \frac{1}{|G|} \sum_{g \in G} \mathbf{1}(\text{pitch}(g) \in C(t_g)).$$

Higher chord-tone ratio indicates stronger harmonic compatibility between generated notes and the reference chord progression.

Register Balance Error. The accompaniment target must distribute note activity across lower and higher pitch ranges in a way that resembles piano left-hand and right-hand usage. Register balance error is computed as the absolute deviation between predicted and reference note proportions in these two ranges. Lower values indicate better agreement with the target texture.

4.4 Comparison Setup

The proposal defined three comparison points: a rule-based baseline, a melody-conditioned Transformer, and a chord-conditioned Transformer. At the milestone stage, the main completed quantitative comparison is between the two neural models.

The rule-based baseline has been implemented and used for qualitative inspection, while the neural comparison is reported quantitatively to isolate the effect of chord conditioning.

5 Milestone Progress

The milestone goals stated in the proposal included completion of preprocessing, implementation of a baseline system, training of the melody-conditioned Transformer, and preliminary evaluation. These goals were achieved. In addition, the chord-conditioned Transformer was also implemented and evaluated within the milestone period.

The project therefore progressed slightly beyond the original milestone plan, because both neural variants in the proposal are already implemented and compared quantitatively.

6 Preliminary Results

Table 3 reports the complete metric comparison between the melody-conditioned and chord-conditioned models.

The chord-conditioned model improves all reported metrics. The increase in chord-tone ratio from 0.4772 to 0.5168 is particularly important because it directly measures harmonic compatibility between the generated notes and the active chord progression. This suggests that explicit harmonic context helps the model choose pitch classes that better align with tonal structure.

The onset-level improvement from 0.1618 to 0.1740 indicates that chord conditioning also improves temporal placement, not just harmonic correctness. At the same time, onset F1 remains substantially higher than note F1 in both models. This pattern suggests that rhythmic placement is learned earlier and more reliably than exact pitch-duration realization.

Register balance error decreases from 0.1333 to 0.1201 in the chord-conditioned model. This indicates that harmonic conditioning also contributes modestly to better global accompaniment texture, especially in the distribution of note activity between lower and higher pitch ranges.

7 Qualitative Observations and Error Analysis

Generated MIDI outputs from both neural models show that the system is learning non-trivial accompaniment structure. The outputs are not empty,

not reduced to a single repeated pitch, and not confined to an unrealistic piano range. This confirms that the pipeline is functioning correctly and that the models are generating structured symbolic content rather than collapsing.

However, several failure modes remain visible.

First, generated accompaniment is often sparser than the reference target. This is consistent with the relatively low note recall values and suggests that the current decoding strategy tends to under-generate notes.

Second, failures are more frequent in passages where the accompaniment is rhythmically dense. When event density increases, the model often misses inner notes or fails to reproduce sustained chordal texture.

Third, the left-hand region remains harder to model than the upper accompaniment range. This is reflected in the register balance error and in qualitative listening, where bass motion and low-register support are often less stable than upper-register chord content.

These observations are consistent with the metric patterns and help explain why onset-level performance is noticeably stronger than full note reconstruction.

8 Future Work

The remaining phase of the project will focus on improving accompaniment prediction quality in ways that directly address the current failure modes.

The first priority is systematic hyperparameter tuning. Planned experiments include adjusting model depth, hidden dimension, number of attention heads, dropout, and phrase length. These changes will be evaluated in terms of note F1, onset F1, and chord-tone ratio.

The second priority is decoding refinement. The current results are obtained using greedy decoding. Beam search will be implemented to allow the model to consider multiple candidate accompaniment continuations at each step. This is expected to improve note recall and reduce locally inconsistent token sequences.

The third priority is representation refinement. The current event-based vocabulary may be too coarse for accurate sustain modeling in dense polyphonic accompaniment. We plan to test finer temporal granularity and alternative token grouping schemes to better capture chord sustain and

Planned milestone item	Status	Evidence
POP909 preprocessing pipeline	Done	track parsing, tokenization, segmentation
Rule-based baseline	Done	generated accompaniment MIDI files
Melody-conditioned Transformer	Done	trained and evaluated on test split
Preliminary evaluation	Done	note/onset/chord/register metrics
Chord-conditioned model	Done	trained and evaluated ahead of plan

Table 2: Milestone progress relative to the proposal plan.

Metric	Mel	Chord
Note Prec.	0.0613	0.0665
Note Recall	0.0791	0.0800
Note F1	0.0676	0.0712
Onset Prec.	0.1452	0.1598
Onset Recall	0.1934	0.2003
Onset F1	0.1618	0.1740
Chord-tone ratio	0.4772	0.5168
LH ratio err.	0.1333	0.1201
RH ratio err.	0.1333	0.1201

Ziyu Wang. 2020. Pop909. In *ISMIR*.

Table 3: Preliminary quantitative results on the test set.

rhythmic density.

The final report will also include deeper qualitative analysis. This will involve piano-roll visualization, comparison against rule-based baseline outputs, and discussion of failure cases across different musical phrases.

9 Conclusion

This milestone extends the proposal into a complete experimental pipeline for melody-conditioned piano accompaniment generation. The preprocessing pipeline, accompaniment baseline, neural models, and evaluation metrics are all implemented. Preliminary experiments show that explicit chord conditioning improves note prediction, rhythmic alignment, harmonic compatibility, and register balance. While exact polyphonic realization remains difficult, the milestone results establish that the approach is feasible and provide a clear experimental direction for the final phase.

References

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